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The critical role of trust in adopting AI-powered educational technology for learning: An instrument for measuring student perceptions

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ABSTRACT

In recent decades, we have witnessed the democratization of AI-powered Educational Technology (AI-EdTech). However, despite the increased accessibility and evolving technological capabilities, its adoption is accompanied by significant challenges, predominantly rooted in social and psychological aspects. At the same time, limited research has been conducted on human factors, especially trust, influencing students' readiness and willingness to adopt AI-EdTech. This study aims to bridge this gap by addressing the multidimensional nature of trust and developing a new instrument for measuring students' perceptions of adopting AI-EdTech. With 665 student responses, we employ Exploratory and Confirmatory Factor Analysis to provide evidence of the instrument's internal validity and identify four key factors influencing students' trust and readiness to adopt AI-EdTech. We then utilize Structural Equations Modeling to explore the causal relationships among these factors, confirming that students' trust in AI-EdTech positively influences AI-EdTech's perceived usefulness both directly and indirectly through AI-readiness. Finally, we use our instrument to analyze 665 student responses, covering eight courses and Bachelor's and Master's degree programs. Our contribution is two-fold. First, by introducing the empirically validated instrument, we address the need for more consistent and reliable assessments of trust-related factors in student adoption of AI-EdTech. Second, our findings confirm that student demographics, specifically gender and educational background, significantly correlated with their trust perceptions, emphasizing the importance of addressing the specific needs of students with various demographics.

1. Introduction

The 2024 EDUCAUSE Horizon report (Teaching and Learning edition) has recently highlighted the transformative impact of artificial intelligence (AI) on higher education (Pelletier et al., 2024). Indeed, AI-powered Educational Technology (AI-EdTech) offers a wide range of applications, including assessment and evaluation, real-time personalized feedback, immersive adaptive learning experiences, and fostering student motivation and engagement, with significant potential to improve educational outcomes (Pelletier et al., 2024; Kasneci et al., 2023). However, despite its potential, the adoption of AI-EdTech by practitioners (both students and educators) has been met with reluctance, often driven by social and psychological barriers such as a lack of trust, perceived risks, and ethical concerns (Kelly et al., 2023; Yan et al., 2024). Addressing these barriers by studying practitioners' perceptions is essential to unlocking the transformative potential of AI-EdTech in educational contexts.

Over the last decades, theoretical frameworks such as the Technology Acceptance Model (TAM) (Davis, 1989) and the Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh et al., 2003) have been foundational in explaining user acceptance of technology through constructs like perceived usefulness and ease of use. While these models have been widely applied in educational settings (Foroughi et al., 2023; Park, 2009), they often fail to capture the unique characteristics of AI-powered systems. Indeed, traditional educational technologies are typically rule-based and grounded in human pedagogical and psychological expertise, which lends them a degree of transparency and explainability. In contrast, AI-EdTech relies on artificial neural networks trained on vast datasets (Vaswani et al., 2017), making its decision-making processes opaque and raising ethical concerns (Glikson & Woolley, 2020; Kelly et al., 2023). Research has shown that these challenges diminish students' trust in AI-EdTech and negatively impact their willingness to adopt these tools (Cukurova et al., 2020; Cojean et al., 2023). Consequently, trust has emerged as a pivotal factor in AI-EdTech adop-

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tion. In response, scholars have increasingly incorporated trust-related factors into their AI-EdTech adoption models. However, most existing studies adopt one-dimensional trust models and rely on proprietary, study-specific formulations. This lack of standardized methods for measuring trust in AI-EdTech hinders the ability to address trust-related challenges systematically and limits the comparability of findings across studies (Ullman et al., 2021).

In response to this gap, our study introduces a novel instrument to assess students' trust in AI-EdTech, conceptualizing trust as a multidimensional construct. By integrating established trust frameworks, utilizing robust statistical methodology, and identifying the key trust-related factors and their interplay, we aim to provide a tool allowing a more comprehensive understanding of the role of trust in AI-EdTech adoption. From this perspective, the proposed trust instrument has the potential to contribute to advancing methodological rigor in trust-related research of AI-EdTech adoption and offer actionable insights for designing and implementing AI-EdTech that aligns with students' needs and expectations.

2. Literature review

In this section, we first review the promising applications of AI-EdTech (Subsection 2.1) and well-established technology acceptance frameworks and models (Subsection 2.2). Next, we highlight the unique challenges associated with AI-powered technology acceptance, particularly those stemming from intrinsic characteristics of AI, such as its not deterministic and black-box nature (Subsection 2.3). These challenges underscore the critical role of human factors, especially trust, as a key predictor of students' willingness to adopt AI-EdTech. We then explore the complexity of measuring trust, emphasizing its multi-dimensional nature, and review frameworks that model trustworthiness and trust (Subsection 2.4). Finally, we discuss how traditional technology acceptance frameworks have been adapted in various studies to incorporate the trust construct and measure its influence on AI-EdTech adoption and pinpoint key gaps in the current research landscape regarding trust in AI-EdTEch (Subsection 2.5).

2.1. Applications of AI-powered technology in educational settings

AI-EdTech offers significant opportunities to enhance learning experiences through diverse applications (Pelletier et al., 2024; Kasneci et al., 2023; Crompton & Burke, 2023). These technologies can effectively provide students with real-time, evidence-based, personalized feedback to revise constructed responses to open-ended questions and academic essays, improving students' engagement and learning outcomes (Dai et al., 2024; Fleckenstein et al., 2023; Meyer et al., 2024). Moreover, conversational chatbots empowered by generative AI can act as interactive learning companions or virtual teaching assistants, fostering more profound understanding, deeper engagement, and collaboration (Kuhail et al., 2023; Wambsganss et al., 2021). Furthermore, AI enables immersive and experiential learning through simulations, augmented reality environments, and the Metaverse (Ahuja et al., 2023). Besides, AI can aid students in managing their learning activities by tracking progress and knowledge acquisition, prioritizing tasks, and recommending courses and learning activities (Frej et al., 2024). Moreover, AI-powered systems can assist teachers in assessment tasks and in personalizing their pedagogy to meet individual student needs (Celik et al., 2022). However, despite their proven effectiveness, students and teachers are often reluctant to use them, making it critical to consider their perceptions and attitudes toward adopting such technology, particularly the role of trust in shaping their willingness to engage with and rely on AI-powered tools in educational settings (Kelly et al., 2023; Yan et al., 2024).

2.2. Frameworks assessing technology adoption

Over the past two decades, many frameworks assessing technology adoption have been proposed. The most widely used are Davis's Technology Acceptance Model (TAM) and its extension, the Unified Theory of Acceptance and Use of Technology (UTAUT), that explain technology adoption through performance expectancy (aka perceived usefulness), effort expectancy (aka perceived ease of use), and social influence, emphasizing that these are key factors influencing users' behavioral intentions (Davis, 1989; Venkatesh et al., 2003). TAM and UTAUT have been widely used in educational contexts to predict teachers' and students' behavioral intentions toward using EdTech (Foroughi et al., 2023; Saihi et al., 2024; Park, 2009; Wang et al., 2021). For example, in the context of higher education, Kim et al. (2020) confirmed that the perceived usefulness and ease of communication with AI-based teaching assistants (TAs) positively predicted students' intent to use them.

Bandura's Social Cognitive Theory (SCT) and its primary construct, Self-Efficacy (SE) (Bandura, 1977; Schwarzer & Jerusalem, 1995), have also been employed to investigate practitioners' readiness to embrace technology in their professional environments. Later, Computer Self-Efficacy (CSE) has been defined as "an individual's perception of efficacy in performing specific computer-related tasks within the domain of general computing" (Compeau & Higgins, 1995; Marakas et al., 1998). This construct has been applied across various fields, including education. For instance, Park (2009) extended the Technology Acceptance Model (TAM) by integrating self-efficacy and demonstrated that it was the most significant predictor of university students' behavioral intention to use e-learning platforms.

Additionally, the Diffusion of Innovation (DoI) Theory (Rogers et al., 2014), which explains the adoption of technological innovations through five key attributes—relative advantage, complexity, trialability, compatibility, and observability—has also been employed to study students' behavioral intentions to adopt AI-EdTech. For example, Ayanwale and Molefi (2024) extended TAM with DoI and found that students' perceived relative advantages of chatbots and their compatibility (i.e., alignment of chatbot use with existing beliefs, values, and experiences) positively influenced their adoption intention. Interestingly, in this study, TAM's perceived usefulness construct was not found to significantly impact adoption intention, emphasizing the unique explanatory power of DoI attributes in this context.

2.3. Adoption of AI-powered technology

Although the valuable insights offered by the existing theories and instruments (especially TAM and its extensions) are widely recognized for their effectiveness in assessing technology adoption, the swift progression of AI technologies has diminished these models' predictive accuracy (Kelly et al., 2023; Sohn & Kwon, 2020).

Previous research has attributed this reduced predictability of the well-established theories to their lack of explicit addressing of the unique aspects of AI-powered tools, precisely their "AI nature" (Glikson & Woolley, 2020; Gursoy et al., 2019; Kelly et al., 2023; Sohn & Kwon, 2020). In contrast to traditional EdTech, AI-EdTech focuses on computer systems capable of accomplishing complex tasks and activities that have historically relied on human cognition, such as natural language processing, problem-solving, logical reasoning, and critical thinking (Pelletier et al., 2024; Dwivedi et al., 2023). From a technological point of view, unlike traditional educational technology, which primarily relies on deterministic systems based on carefully curated expert knowledge, AI-EdTech is built on artificial neural networks, probabilistic models, and a huge amount of training data (Vaswani et al., 2017). Such sophisticated architecture allows it to mimic human behavior and dynamically adapt and improve through continuous interaction with learners and learning environments. However, it also results in opaque decision-making processes and a lack of justification for the resulting insights. Consequently, learners and educators struggle to interpret the rationale behind AI-driven recommendations and often cannot assess their validity and reliability (Glikson & Woolley, 2020; Dwivedi et al., 2023).

Evidence across various disciplines highlights that the non-deterministic nature, coupled with a lack of transparency and explainability, poses significant challenges for users (Kelly et al., 2023; Dwivedi et al., 2023). Such issues influence users' trust and adversely affect practitioners' perceptions and attitudes toward adopting such tools, ultimately hindering the widespread adoption of AI-powered technologies (Cukurova et al., 2020; Cojean et al., 2023; Nazaretsky et al., 2021; Ragot et al., 2020; Pinto dos Santos et al., 2019; Gruenhagen et al., 2024). For example, in the field of medical radiology, where AI has been shown to surpass human performance (Rajpurkar et al., 2017), existing studies have indicated that experts tend to view recommendations less favorably when they are presented as originating from AI, as opposed to those same recommendations being attributed to human sources (Gaube et al., 2021). Cukurova et al. (2020) found that framing educational research evidence within AI reduced its general public's perceived credibility compared to framing it within neuroscience or educational psychology, highlighting reduced trust in AI-based technology. This oversight of trust's influence may account for the reduced applicability of well-established technology acceptance models to AI-powered technology. This underscores the importance of incorporating trust-specific factors to analyze its adoption processes effectively (Kelly et al., 2023).

2.4. Frameworks for measuring trustworthiness and trust

Before discussing frameworks and models for measuring trust, it is crucial to establish clear definitions and assumptions.

First, in this study, we adopt the definition of trust provided by Lee and See (2004): "Trust can be defined as the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability." According to this definition, trust can only be measured in scenarios that involve *uncertainty*, *vulnerability*, and *high expectations* (Lee & See, 2004; Vereschak et al., 2021; Mayer et al., 1995). In educational settings, particularly with AI-EdTech, these conditions are prevalent. Research shows that students often hold *high expectations* for the potential benefits of AI-EdTech (Lünich et al., 2024; Gruenhagen et al., 2024; UNESCO et al., 2024b; Ayanwale & Molefi, 2024). At the same time, the high stakes involved, such as future educational and career opportunities, amplify student *vulnerabilities* (Lünich et al., 2024). Moreover, AI technology's black-box characteristics complicate predicting and understanding their behaviors and outcomes. This inherent opacity, combined with the complexities of educational settings, introduces substantial *uncertainties*. Together, these factors make a trust-centered approach applicable to studying AI-EdTech adoption.

Second, trust is a function of the trustee (the entity being trusted, in this case, AI-EdTech) and the truster (the individual placing trust in the trustee, in this case, student) characteristics. So, it is essential to distinguish between trustworthiness — a property of the trustee — and trust, which reflects the truster's attitude. Both, in this case, must be considered within the specific context of learning. Furthermore, surveys relying on self-reported data do not measure objective trustworthiness and real trust; instead, they capture the truster's *perceptions* of the trustee's trustworthiness and their *intent* to trust.

Trustee characteristics - AI-EdTech trustworthiness. Previous research in human-computer interaction adopted several frameworks for measuring trust's role in human-human and human-organization relationships. The frameworks such as the Muenster Epistemic Trustworthiness Inventory (METI) (Hendriks et al., 2015), Model of Organizational Trust (Mayer et al., 1995), Human-Computer Trust (HCT) Model (Hoff & Bashir, 2015) and Multi-Dimensional Measure of Trust (MDMT) (Malle & Ullman, 2021) mostly focus on *the trustee's trustworthiness*. They all distinguish between *performance* and *moral* dimensions of the trustworthiness. The performance dimension is defined by the following characteristics: competence, validity, usefulness, helpfulness, ability, skills, expertise, as well as reliability and predictability. It is also equivalent to the TAM's usefulness and UTAUT's performance expectancy dimensions. The moral dimension is aimed at assessing if the trustee is aligned with human values and moral norms, and it is often divided into two or three sub-dimensions. The most common are *transparency* and *integrity*. Transparency refers to the extent to which the trustee explains or justifies their behavior, or the trustee's internal logic is accessible to the truster. On the other hand, integrity refers to the perception that the trustee is motivated by goodwill and intends to act honestly to benefit and protect the truster. It is often defined by one of the following terms: benevolent, sincere, honest, just, fair, moral, and ethical.

Truster characteristics - students' readiness and intent to trust. It has been shown in the educational domain that increasing users' familiarity and epistemic knowledge about AI-powered technology influence their intent to trust it (Hoff & Bashir, 2015; Touretzky et al., 2019; Luckin et al., 2022; Nazaretsky et al., 2022a). This refers to a distinct set of characteristics predominantly centered around AI literacy and AI readiness (Laupichler et al., 2022; Long & Magerko, 2020; Luckin et al., 2022; Ng et al., 2021). Long and Magerko (2020) defines AI literacy as "a set of competencies that enables individuals to critically evaluate AI technologies; communicate and collaborate effectively with AI; and use AI as a tool online, at home, and in the workplace." Among others, they specify the requirement of knowing that AI learns from training data using techniques chosen by its developers, and its results should be critically evaluated and interpreted for their validity in the specific context of the application before using them. In the context of truster's readiness, (Hoff & Bashir, 2015; Lee & See, 2004) suggested including self-efficacy (SE) (Bandura, 1977) and close to it self-confidence. Moreover, the Resistance to Organizational Change Theory exploring employees' willingness to accept change (Erwin & Garman, 2010; Oreg, 2006) proposes, in addition to SE, to consider potential threats such as loss of autonomy, change in existing routines and apprehensions about lacking the necessary skills to perform effectively in the altered environment.

Finally, research confirms that individual truster characteristics, such as demographics (e.g., age, gender, level of education) (Strzelecki & ElArabawy, 2024), along with dispositional traits like propensity to trust (Hoff & Bashir, 2015; Mayer et al., 1995) and predisposition to resist and openness to change (Erwin & Garman, 2010; Oreg, 2006), also play a significant role. The dispositional traits reflect the inherent personality characteristics of the truster, independent of the trustee's specific attributes or the context.

2.5. Role of trust in adopting AI-EdTech

The acceptance of AI technologies is strongly influenced by the specific contexts in which they are implemented, as demonstrated by studies across various fields (Kelly et al., 2023; Luo et al., 2010; Sheng et al., 2008). In the educational domain, there has been a notable increase in research examining practitioners' attitudes toward adopting AI-EdTech, with more than a thousand studies published within the last year alone.[1] However, despite the recognized importance of trust, only a small fraction of these studies have incorporated it into their models,[2] highlighting that trust remains an underexplored aspect in the field of adoption AI-EdTech.

A careful review of the above results revealed that most studies addressing the topic used TAM and added a single trust-related item or single trust-related factor represented by several items to their model. A single item was usually formulated in general terms like "I trust the information provided by ChatGPT", "The more I used Bing Chat, the more

[1] Based on a SCOPUS database search of article titles, abstracts, and keywords using the query ("Education*" AND ("Artificial Intelligence" OR "AI") AND ("learn*" OR "student*") AND ("attitude*" OR "perception*")).

[2] Adding "Trust" to the search criteria reduced the results to only 5% of the initial set.

I felt that I could rely on this tool if I needed it" or "ChatGPT is an unreliable source of knowledge - I do not trust it" (Kamoun et al., 2024; Bubaš et al., 2023; Bikanga Ada, 2024). A single factor was usually related to a lack of information security and possible privacy violations (Rahman et al., 2022; Maheshwari, 2024; Jo, 2023; Saihi et al., 2024) or lack of accuracy (Bego et al., 2024). The rest, with only a few exceptions, explored two or more trust-related factors (typically, risks-related or general AI-trustworthiness) by one item each (Ayanwale & Molefi, 2024; Choi et al., 2023; Lai et al., 2024; Foroughi et al., 2024; Bego et al., 2024; Duong, 2024; Polyportis & Pahos, 2024; Almogren et al., 2024; Bernabei et al., 2023). Notably, all the above studies underscores the importance of incorporating trust-related items into their models for understanding students adoption of AI-EdTech.

Interestingly, most of these studies, published between 2023 and 2024, focused on conversational chatbots like ChatGPT. This reflects the rapid adoption of this technology in educational settings, accompanied by concerns about its use. On the positive side, the rapid adoption of AI-EdTech has prompted scholars to deepen their understanding of trust and other human factors in technology adoption. However, these single-item and single-factor approaches were limited because they failed to capture the complexity and multidimensionality of trust. And even more importantly, these studies had no consensus on which aspects to include and how to formulate the corresponding trust-related items. Nearly half of these studies used self-created items proprietary to their research, while the rest reused trust-related items from studies in domains other than educational, with little to no overlap between them. Among items crafted for the specific studies were those highly related to the study context (Bego et al., 2024) or, on the opposite, too general. Among the few that adopted multi-dimensional trust models similar to MDMT (Malle & Ullman, 2021) are Nazaretsky et al. (2022b), who proposed an instrument for measuring teacher's trust in AI-EdTech and validated it based on data collected from 132 science teachers, Tossell et al. (2024) who evaluated the change in 24 students trust after intervention involved using ChatGPT for essay writing. In addition, Pitts et al. (2024) proposed a multi-dimensional model for measuring students' acceptance and trust of Conversational AI for learning. However, they did not provide empirical evidence of the corresponding instrument's internal validity.

This lack of uniformity makes it challenging, if not impossible, to compare or replicate the results of these studies. This situation is not rare in the Human-Computer Interaction field (Ullman et al., 2021; Open Science Collaboration, 2015; Hoffman & Zhao, 2020) and may, over time, reduce the credibility of the results within the scientific community and the general public.

3. The present study

In response to the identified importance of introducing trust into student AI-EdTech adoption models and the lack of uniformity in measuring trust within AI-EdTech, our study presents a comprehensive survey instrument that conceptualizes trust as a multidimensional construct. By developing this instrument, we aim to promote the comparability and replicability of future trust-related research on AI-EdTech adoption. Our goal is to cover various dimensions of trust, including the performance and moral trustworthiness of AI-EdTech as an information source, students' readiness to adopt AI for learning, and their intent to trust it, providing a more nuanced understanding of students' trust in AI-EdTech.

To accomplish this, we conducted a study that consisted of two phases. The first aims to develop instrument items based on the established multidimensional trust models, utilize Exploratory and Confirmatory factor analysis to provide empirical evidence for its internal validity, and explore the factors influencing students' trust in AI-EdTech. In addition, this phase aims to investigate the interplay between the emerging factors using Structural Equation Modeling. The goal of the second phase is to use the instrument to portray higher education students' trust-related perceptions.

4. Methodology

Our procedure consisted of two main phases. In the first phase, we developed the instrument items and administered them to a population of 665 students to validate them. In the second phase, we analyzed students' responses in detail.

Procedure. Our two-phase procedure illustrated in Fig. 1 closely followed the instrument development methodology suggested by Butz and Branchaw (2020).

In the first phase (Section 5, Instrument Development), we developed and evaluated our instrument in five steps. Step 1 consisted of developing items and piloting the instrument. We then administered the instrument to 665 students.

Next, we performed an Exploratory and Confirmatory factor analysis of student survey responses. To this end, we divided the collected responses into two parts. In step 2, we used an Exploratory Factor Analysis (EFA) on one part of the data (373 students) to provide evidence for our proposed instrument's internal factorial structure. In step 3, we validated the internal consistency of the emerged factors and labeled them. Next, in step 4, we continued with a Confirmatory Factor Analysis (CFA) on the other part of the response data (292 students) to further confirm the internal factorial structure of the proposed instrument. Finally, in step 5, we investigated the causal relations of the emerged factors using Structured Equation Modeling (SEM).

In the second phase (Section 6, Responses Analysis), we analyzed all student responses to the proposed instrument both quantitatively and qualitatively to gain insights into the influence of students' demographics on facilitators and barriers to adopting AI-EdTech for learning.

Population. We administered the instrument to 665 students at EPFL. Students were registered for eight courses from different programs (Computer Science, Communication Systems, Mathematics, Physics, Life Science, Mechanical, and Civil Engineering) offered at the Bachelor's and Master's levels. The courses covered diverse STEM topics, such as Data Analysis, Machine Learning, Discrete Mathematics, Calculus, and Python and C++ Programming. The study was approved by the university's human research ethical committee (HREC 017-2023). Participation in the research study was voluntary and restricted to students of legal age. Participants completed the survey online (10-15 minutes). 373 students attended classes taught in English and 292 in French.[3] Participants' age varied from 18 to 38 (μ = 19.7, median = 19, σ = 2.35). Table 1 presents the detailed student demographics.

5. Instrument development

We developed our instrument for measuring trust in five steps (see Fig. 1). In the following, we describe each step in detail.

Step 1. Instrument Items Development. The instrument items were constructed using a top-down approach based on existing models for measuring trust as a multidimensional construct (Subsection 2.4), previous research in adopting AI-EdTech (Subsection 2.5), teachers' trust instrument introduced in Nazaretsky et al. (2022b) and discussions with educational technology practitioners. The initial draft of the instrument consisted of three parts: i) Benefits of using AI in educational settings, ii) Potential risks and obstacles involved in using AI in educational settings, and iii) Working alongside AI in educational settings. The first and second parts covered students' perceptions of AI trustworthiness. More specifically, the first part explored the MDMT performance dimension by naming possible applications of AI in educational settings, and the second part explored the MDMT moral dimension by naming possible

[3] Some first-year Bachelor courses in EPFL are taught in French. All other courses are in English.

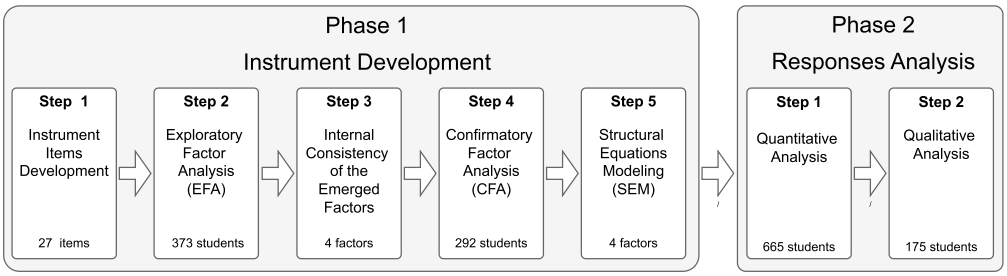

Fig. 1. Overview of our instrument development procedure. We first developed and validated a student perceptions survey in five steps (Phase 1, Section 5) and then analyzed the students' responses by quantitative and qualitative analysis (Phase 2, Section 6).

Table 1
Descriptive statistics per language and the entire sample, including absolute counts and percentages (in brackets).

	Group	English (N = 373)	French (N = 292)	Total (N = 665)
Age	18 - 19	183 (49%)	225 (77%)	408 (61%)
	20 - 21	85 (23%)	55 (19%)	140 (21%)
	22 - 23	65 (17%)	5 (<2%)	70 (11%)
	≥ 24	40 (11%)	7 (2%)	47 (7%)
Gender	Male	280 (75%)	197 (67%)	477 (72%)
	Female	86 (23%)	89 (30%)	175 (26%)
	Not-Binary	5 (< 2%)	2 (< 1%)	7 (1%)
	Other	2 (< 1%)	4 (< 2%)	6 (< 1%)
Degree	Bachelor (1st year)	249 (67%)	292	541 (81%)
	Bachelor (2nd year)	9 (2%)	-	9 (< 2%)
	Master	115 (31%)	-	115 (17%)
Program	Life Science, Mechanical and Civil Engineering (Eng)	9 (2%)	162 (55%)	171 (26%)
	Mathematics, Physics (M&P)	-	130 (45%)	130 (19%)
	Computer Science, Communication Systems (C&C)	364 (98%)	-	364 (55%)

Table 2
MDMT Performance dimension: Perceived Benefits. The question was formulated as follows: Below are some opinions that students have in the field of adopting AI-based tools for learning. The sentences address the potential benefits of using AI in education. Read each sentence carefully and decide how much it characterizes your opinion and feelings. Choose the option that reflects your position.

	Item	References
PB1	AI can assist me in improving my responses (e.g., answers to open-ended questions, writing essays, etc.) by suggesting personalized feedback in real time.	Beigman Klebanov and Madnani (2020); Ramesh and Sanampudi (2022)
PB2	AI can assist me in self-learning by adjusting automatically individualized learning tasks adapted to my personal needs and level of knowledge.	Aleven et al. (2006); Heffernan and Heffernan (2014); Kasneci et al. (2023)
PB3	AI (e.g., ChatGPT) can serve me as a learning pal or teaching assistant.	Kuhail et al. (2023); Wambsganss et al. (2021)
PB4	AI can help teachers diagnose my strengths and difficulties so teachers can assign individualized learning tasks adapted to my personal needs and level of knowledge.	Celik et al. (2022); Nazaretsky et al. (2022b)
PB5	AI can assist me with managing my learning activities (e.g., identifying if I'm on track with my tasks, offering task prioritization, optimizing time management, etc.)	Jin et al. (2023); Crompton and Burke (2023)
PB6	AI can assist me in preparing for exams by identifying my knowledge gaps and providing relevant learning materials (e.g., explanations, tasks, etc.)	Kasneci et al. (2023); Crompton and Burke (2023)
PB7	AI can support learning simulations and augmented reality environments for learning (e.g., flight simulator).	Ahuja et al. (2023)

concerns and risks involved. The third part covered the student's intent to trust dimension, which consisted of items related to students' AI readiness and AI literacy. Below, we present the process in more detail.

Benefits of Using AI in educational settings.

For the MDMT performance dimension, we constructed seven items (PB1-PB7) based on the common applications of AI-powered technology for learning and teaching (Subsection 2.1). The items are presented in Table 2.

Potential Risks and Obstacles.

For the MDMT moral dimension, we formulated ten items PO1-PO10 (Subsection 2.4). Items PO4-PO7 explored the transparency sub-dimension focusing on uncertainty about how AI-EdTech reasons and makes decisions, what datasets are used to create AI models, who is accountable for AI decisions, and the AI tools creators' intentions and funding. The rest explored the integrity sub-dimension with PO1-PO3 considering the AI-EdTech lack of human characteristics and PO8-PO10 ethical aspects as possible biases, lack of diversity, AI mistakes, ownership of what data is collected, data breaches, mass surveillance, and profiling. The items are presented in Table 3.

Working alongside AI in Educational Settings.

To explore truster (student in this case) characteristics, we formulated ten items (Subsection 2.4). Specifically, four items PR1-PR4 explore students' readiness to use AI for learning, considering their self-efficacy and mastering required skills (Table 4). In addition, six items PT1-PT6 discuss student intent to trust AI. The items are presented in Table 4.

Table 3
MDMT Moral dimension: Perceived Obstacles. The question was formulated as follows: Below are some opinions that students have in the field of adopting AI-based tools for learning. The sentences address the potential obstacles related to using AI in education. Read each sentence carefully and decide how much it characterizes your opinion and feelings. Choose the option that reflects your position.

	Item	References
PO1	AI developers lack educational experience.	Nazaretsky et al. (2022b)
PO2	AI algorithms do not understand social, emotional, and motivational factors that are very important in learning and teaching.	Nazaretsky et al. (2022b)
PO3	AI algorithms do not know my personal learning history outside the system.	Nazaretsky et al. (2022b)
PO4	AI's decision-making and reasoning processes are not transparent.	Long and Magerko (2020); Ng et al. (2021)
PO5	The origins, collection methods, and limitations of the data utilized by AI are not clear (e.g., who created the dataset, how the data was collected, etc.)	Long and Magerko (2020); Ng et al. (2021)
PO6	It's unclear who holds accountability for decisions made by AI.	Long and Magerko (2020); Ng et al. (2021)
PO7	The intentions and funding sources of AI-based technology providers are not transparent.	Long and Magerko (2020); Ng et al. (2021)
PO8	It's unclear when and what type of data AI gathers about me.	Long and Magerko (2020); Drachsler and Greller (2016)
PO9	The use of data raises risks to my privacy (e.g., data breaches, surveillance and profiling, etc.)	Long and Magerko (2020); Drachsler and Greller (2016)
PO10	The use of data raises ethical issues (e.g., possible biases, lack of diversity, AI mistakes, etc.)	Long and Magerko (2020); Drachsler and Greller (2016)

Table 4
AI Readiness and Trust Intent Dimensions: working alongside AI in Educational settings. The question was formulated as follows: Read each sentence carefully and decide how much it characterizes your opinion and feelings. Choose the option that reflects your position.

	Item	References
PR1	I believe I will be successful in using AI-based tools to personalize my learning when and if they are available.	Bandura (1977); Oreg (2006)
PR2	I will use AI-based tools for personalizing my learning when and if they are available.	Oreg (2006); Nazaretsky et al. (2022b)
PR3	I am sure that using AI-based tools for personalizing my learning will require me to change my learning habits significantly.	Oreg (2006); Nazaretsky et al. (2022b)
PR4	I am sure that using AI-based tools for personalizing my learning will require me to gain new skills (e.g., awareness of and dealing with possible AI mistakes, biases, critical and ethical consuming of AI insights, etc.)	Oreg (2006); Nazaretsky et al. (2022b); Long and Magerko (2020)
PT1	As AI-based tools for personalized learning become more prevalent, I will trust them more.	Nazaretsky et al. (2022b)
PT2	The more I know how an AI-based tool makes decisions (e.g., the steps involved in machine learning and the practices and challenges that each step entails, etc.), the more I can trust it.	Long and Magerko (2020); Ng et al. (2021); Nazaretsky et al. (2022b)
PT3	The more data is available for AI-based tools, the more I can trust its insights.	Long and Magerko (2020); Nazaretsky et al. (2022b)
PT4	The more autonomy AI-based tools provide me/my teacher to check/overwrite its automated decisions, the more I can trust them.	Long and Magerko (2020); Nazaretsky et al. (2022b); Dietvorst et al. (2018)
PT5	I rely on the recommendation of an AI-based tool at least as much as I trust a recommendation from a fellow learner.	Nazaretsky et al. (2022b); Dietvorst et al. (2015)
PT6	I rely on the recommendation of an AI-based tool at least as much as I trust a recommendation from my teacher.	Nazaretsky et al. (2022b); Dietvorst et al. (2015)

To assess the proposed instruments' content and face validity, we piloted it with a focus group of four educational experts with more than ten years of research experience in EdTech and learning sciences and ten university students. The focus group was instructed to complete the survey and evaluate it based on the following aspects: i) The survey covers all relevant aspects of adopting AI-EdTech for learning; ii) The instructions for completing the survey and the survey items are articulated and tailored to the language comprehension of the target audience; and iii) The time duration required to complete the survey. Based on the focus group's written responses, we added refining examples and reformulated several items to avoid ambiguities. The final proposed instrument consists of 27 5-level scale Likert items divided into three categories and several open-ended questions soliciting participants' additional considerations and gathering socio-demographic information.

The instrument items were initially developed in English and then translated into French by the fourth author. The translation was validated by two additional researchers, experts in AI-EdTech, and using the back-translation to English technique. The complete instrument is available in our public repository https://github.com/tanya-nazaretsky/trust.

Step 2. Exploratory Factor Analysis. To provide evidence for the proposed instrument's internal structure, we used the responses to the English version of the survey ($N = 373$) and conducted an EFA. We detected multivariate outliers using the Mahalanobis distance and consequently removed 28 students from the analysis, performing the EFA on 345 students. The Kaiser-Meyer-Olkin measure of sampling adequacy ($KMO = 0.76 > 0.7$) and Bartlett's test of sphericity ($\chi^2 = 2076.94$, $p < .0001$) indicated that the collected data was suitable for performing an adequate EFA (Kline, 2013). A Shapiro-Wilk test ($p < .05$ for all items) and Mardia's kurtosis (7.88) and skewness (4758.04) tests ($p < .001$) confirmed that the data was not normally distributed. There was no missing data, as all Likert items were mandatory.

We performed the EFA using a principal axes estimator[4] in R, which is suitable for handling ordinal (Likert) scales and non-normal distribution (Revelle, 2017). A Horn's Parallel Analysis (Horn, 1965) suggested a five-factor structure. To estimate the goodness of fit of the proposed model, we used multiple fit measures simultaneously (American Educational Research Association, 2018; Brown, 2015; Kline, 2013): the root mean square error of approximations RMSEA $= 0.037 \leq 0.05$ with a 90% CI of $[0.029, 0.046] \leq 0.05$, the root mean square of the residuals SRMR $= 0.035 \leq 0.05$, Tucker Lewis Index of factoring reliability TLI $= 0.9 \geq 0.9$ and $\chi^2/df = 335.32/345 < 1$. We used a threshold of $\tau > 0.30$ for the factor loadings to decide on the mappings of items to factors. The resulting factors and items, including the factor loadings, are presented in Fig. 2. The entire table of factor loadings and proportion of variance explained is presented in Appendix Table A.9.

Step 3. Internal Consistency of the Emerged Factors. We examined the internal consistency of the proposed factors using Cronbach's α. We evaluated the corresponding factors' composition with and without the

[4] *fa* component of *psych* R package.

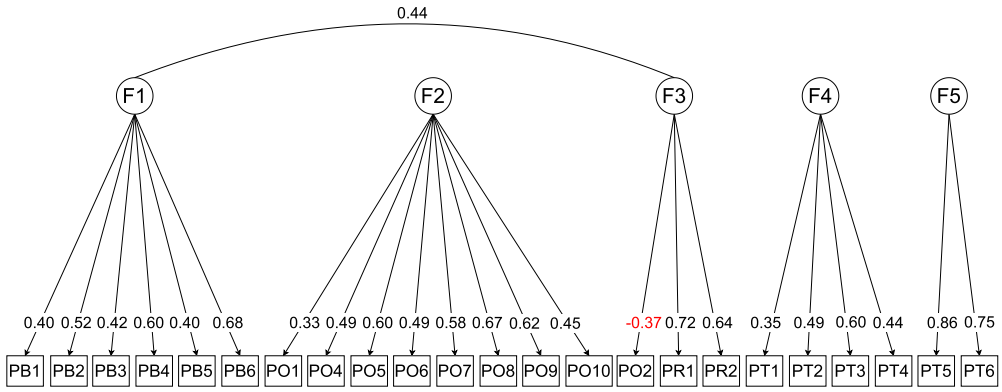

Fig. 2. 5-factor EFA model with a cut-off of 0.30 for factor loadings. The items are represented by boxes, and the factors (F1-F5) by ellipses. Factor F1 correlates with Factor F3. Positive loadings are presented in black, and negative loadings are in red. The four items with no loadings > 0.30 (PB7, PO3, PR3, and PR4) are omitted from the diagram. There are no items with double loading.

Table 5
Reliability analysis of factors with their corresponding Cronbach's α values. Factor F3 consisted of three items: PO2, PR1, and PR2. However, removing item PO2 (with a low negative loading of −0.37) from factor F3 yielded improvement from $\alpha = 0.60$ to 0.77. Similarly, removing item PO1 (with a low loading of 0.33) from factor F2 yielded improvement from $\alpha = 0.75$ to 0.76. Factor F5 consisted of two items only, while factor F4 had a relatively low $\alpha = 0.61$. As F4 and F5 were slightly correlated, we combined them into one factor, yielding an improvement from $\alpha = 0.61$ to 0.66.

Factor	Items	Cronbach's α	Name
F1	PB1 - PB6	0.74	*Perceived Usefulness*
F2	PO4 - PO10	0.76	*Perceived Obstacles*
F3	PR1, PR2	0.77	*Perceived Readiness*
F4	PT1 - PT4	0.61	
F5	PT5, PT6	0.79	
F4 + F5	PT1 - PT6	0.66	*Perceived Trust*

items with relatively low loadings $0.30 < \alpha < 0.4$ (PO1, PO2, PT1) to check their influence on Cronbach's α values of the corresponding factors. In addition, we tried to combine the factors loaded by two items only (F3 and F5) with other correlated factors (Appendix, Table A.10). The resulting Cronbach's α values for the factors and the factor names are presented in Table 5. We obtained in total four factors, with three of them (F1, F2, F3) yielding an adequate α (between $0.7 - 0.8$ (Hair et al., 2009)). The fourth factor (F4 + F5) resulted in $\alpha = 0.66$ (between $0.6 - 0.7$). Though it did not achieve the threshold of 0.7, it can be considered acceptable due to the study's exploratory nature (in particular, all corresponding item loadings on the factor were acceptable) and its complex psychological nature (Kline, 2013).

We then analyzed the emerging factors to determine whether they represented stand-alone properties aligned with the instrument's top-down categories: Items PB1-PB6 (**F1**) discuss the possible benefits of using AI-EdTech for learning. We hence named it *Perceived Usefulness*. Items PO4-PO10 (**F2**) are related to possible obstacles, concerns, and risks in using AI-EdTech for learning. We therefore denoted the factor with *Perceived Obstacles*. Items PR1 and PR2 (**F3**) discuss students' confidence and willingness to adopt AI-EdTech for learning, leading to the name *Perceived Readiness*. Finally, items PT1-PT6 (**F4 + F5**) are related to students' trust in AI-EdTech in various scenarios and situations. So we called the factor *Perceived Trust*.

To summarize, our analysis yielded 21 items divided into four reliable factors *Perceived Usefulness* (6 items), *Perceived Obstacles* (7 items), *Perceived Readiness* (2 items), and *Perceived Trust* (6 items).

Step 4. Confirmatory Factor Analysis. To confirm the proposed instrument's internal structure, we conducted a CFA on the responses to the French version of the survey ($N = 292$). First, we detected multivariate outliers using the Mahalanobis distance, removing 22 students from the analysis. A Shapiro-Wilk test ($p < .05$ for all items) and Mardia's kurtosis (4.31) and skewness (4426.8) tests ($p < .001$) confirmed that the data was not normally distributed.

We performed the CFA using a weighted least square mean and variance adjusted (WLSMV) estimator[5] in R, which is suitable for handling ordinal (Likert) scales and non-normal distributions (Schumacker & Lomax, 2004). Initially, we evaluated the goodness of fit of the 4-factor model emerging from EFA (see Table 5) based on the following measures and thresholds simultaneously (Schumacker & Lomax, 2004): the root mean square error of approximations RMSEA ≤ 0.05, the root mean square of the residuals SRMR ≤ 0.08, Tucker Lewis Index of factoring reliability TLI ≥ 0.9, Comparative Fit Index CFI ≥ 0.95, and $\chi^2/df < 2$. The results (RMSEA $= 0.083 > 0.05$, SRMR $= 0.088 > 0.08$, TLI $= 0.902 > 0.9$, CFI $= 0.915 < 0.95$, and $\chi^2/df = 532.54/183 < 2$) suggested that although the initial model had some aspects aligning well with the data, some areas did not represent the data adequately. As per individual item loading on the proposed factors, all 21 items had significant loading (> 0.30) on the corresponding factors. However, item PT6 had a loading equal to 0.33 (Appendix, Table A.11, 'Initial Model' column), and we excluded it from our further analyses.

Next, we conducted an Index Modification analysis to improve the model's specification. In particular, the analysis suggested to consider the correlations between the following pairs of items: PO5 with PO4 and PO6, PT2 with PT3 and PT4, PT3 and PT4. The improved model taking into account the above pairs of correlated items, indicated a strong fit to the data with RMSEA $= 0.050 \leq 0.05$ and a 90% CI of $[0.039, 0.060] \leq 0.08$, SRMR $= 0.068 \leq 0.08$, TLI $= 0.966 > 0.9$, CFI $= 0.972 > 0.95$, and a $\chi^2/df = 266.23/159 < 2$. All 20 items have significant loading (> 0.30) on the corresponding factors (Appendix, Table A.11, 'Final CFA Model' column). It is essential to mention that although the *Perceived Readiness* factor loaded on two items only (PR1, PR2), these items demonstrated exceptionally high statistically significant ($p < .001$) factor loadings of 0.88 and 0.82, respectively, indicating a strong association with the underlying factor. This result suggests that both items are excellent indicators of the *Perceived Readiness* factor.

The covariance analysis (see Table 6) within our CFA model revealed meaningful insights into the interrelations among the proposed factors. Notably, *Perceived Usefulness* demonstrated a significant positive relationship with both *Perceived Readiness* and *Perceived Trust*. We also found a strong positive relation between the *Perceived Trust* and *Perceived Readiness* factors. While the *Perceived Obstacles* factor showed a

[5] *cfa* component of *lavaan* R package.

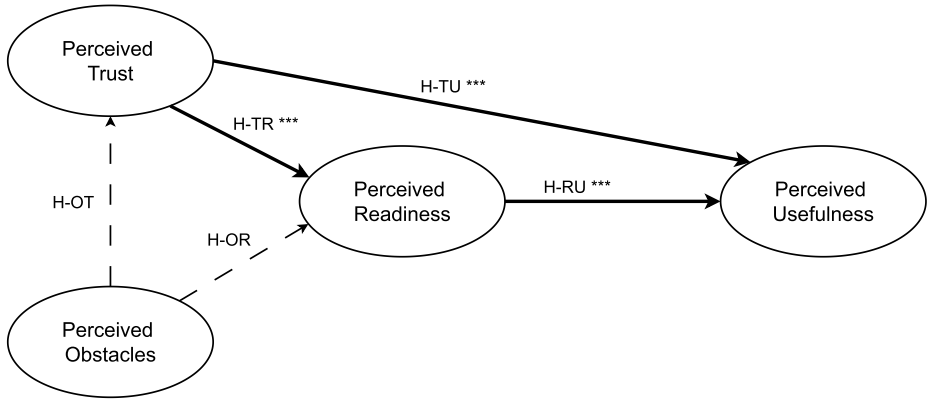

Fig. 3. SEM Analysis. Hypotheses of causal relations with positive relations represented by solid line arrows and negative relations by dashed line arrows. The direction of the arrow indicate the direction of the relation. The significant relationships are represented by bold lines. Significance levels: *** $p < .001$.

Table 6
Correlations between factors of the final 4-factor CFA model. Statistically significant correlations are in bold. Significance levels: *** $p < .001$; ** $p < .01$.

	Perceived Obstacles	*Perceived Readiness*	*Perceived Trust*
Perceived Usefulness	0.09	**0.75*** **	**0.70** **
Perceived Obstacles		**-0.21** **	-0.06
Perceived Readiness			**0.65*** **

Table 7
Effect estimates (with standard errors in brackets) for the proposed SEM model. Statistically significant relationships are in bold.

Hypothesis	Estimate	Standardized Estimate	p-value
H-TU	**0.76 (0.10)**	**0.45 (0.04)**	.0006
H-TR	**0.74 (0.08)**	**0.58 (0.04)**	< .0001
H-RU	**0.58 (0.04)**	**0.44 (0.03)**	< .0001
H-OR	-0.16 (0.05)	-0.13 (0.03)	.14
H-OT	-0.09 (0.05)	-0.09 (0.05)	.43

weak but significant negative relationship with the *Perceived Readiness* factor. We used these insights as a basis for the SEM.

Step 5. Structural Equation Modeling. We based the SEM hypotheses on our CFA results as well as findings from previous research. Prior research confirms that practitioners' trust in technology positively influences their perceived usefulness of technology-based tools, directly as well as indirectly, through their readiness to utilize them for working purposes (Kelly et al., 2023; Venkatesh et al., 2003; Wu et al., 2011). Furthermore, in the context of AI, Choung et al. (2023) explored the role of trust in adopting AI technologies and found that trust significantly influences usage intention via perceived usefulness. Moreover, anxieties and negative feelings can significantly reduce teachers' trust and readiness to employ AI-based technology (Mac Callum et al., 2014; Park et al., 2019; Wang et al., 2021). We therefore made the following hypotheses (illustrated in Fig. 3):

H-TU *Perceived Trust* positively influences *Perceived Usefulness*.
H-TR *Perceived Trust* positively influences *Perceived Readiness*.
H-RU *Perceived Readiness* positively influences *Perceived Usefulness*.
H-OR *Perceived Obstacles* negatively influence *Perceived Readiness*.
H-OT *Perceived Obstacles* negatively influence *Perceived Trust*.

We employed a 10-fold cross-validation on the responses used for the CFA ($N = 270$) to evaluate the proposed model. Specifically, we tested the model for its ability to predict outcomes consistently across different data subsets. The proposed measurement model demonstrated high consistency across all folds, with fit indices of CFI $= 0.97 > 0.95$, TLI $= 0.96 > 0.9$, RMSEA $= 0.05 \leq 0.05$, and SRMR $= 0.07 < 0.08$. The resulting estimates and standardized effect estimates (averaged over 10 folds), along with significance, are presented in Table 7. We observe that the structural model assessment identified both significant (H-TU, H-TR, and H-RU) and insignificant (H-OT and H-OR) relations. Consequently, we rejected the hypotheses H-OT and H-OR. For the hypotheses H-TU, H-TR, and H-RU, our results suggest that the proposed model is robust with stable significant predictive capacity across different data subsets. Therefore, we accepted the hypotheses H-TU, H-TR and H-RU. The final resulting SEM model is highlighted in Fig. 3 by the bold arrows.

6. Responses analyses

In *Phase 2* (see Fig. 1), we evaluated the responses of all 665 students to gain insights into facilitators and barriers to the adoption of AI-EdTech for learning. We performed a quantitative analysis of the Likert-scale items before qualitatively analyzing students' responses to the open-ended questions.

6.1. Quantitative analysis

We analyzed the overall score distribution per item and factor before investigating the differences between demographics. Fig. 4 illustrates the score distribution for each factor. The per-item results are presented in the Fig. A.5. We observe that students acknowledged the usefulness of AI-EdTech for learning (*Perceived Usefulness*: $\mu(\sigma) = 3.62$ (0.68)), reflecting an overall positive outlook on the potential benefits of implementing AI in educational settings. Moreover, participants showed high intent and substantial confidence in their ability to effectively use AI-EdTech (*Perceived Readiness* $\mu(\sigma) = 3.74$ (0.86)). It is also promising that the importance attributed to mastering AI literacy competencies (item PR4) was relatively high among participants ($\mu(\sigma) = 3.93$ (0.83)), revealing a consensus on the necessity of understanding AI to effectively utilize AI-EdTech. Our study also uncovered several student concerns that could potentially hinder the adoption of AI-EdTech. These concerns revolve around lack of transparency, privacy violations, and the potential for inducing anxieties (*Perceived Obstacles*: $\mu(\sigma) = 3.79$ (0.60)). The concerns covered AI's capability to comprehend real learning contexts outside the AI systems (item PO3: $\mu(\sigma) = 4.00$ (0.91)) and AI's aptitude in recognizing socio-emotional factors important in the learning process (item PO2: $\mu(\sigma) = 3.75$ (1.07)). Students also clearly preferred teacher recommendations over those proposed by AI-EdTech (item PT6: $\mu(\sigma) = 2.60$ (1.13)). In addition, students rated their trust slightly above the neutral point with moderate variability in their responses (*Perceived Trust*: $\mu(\sigma) = 3.31$ (0.66)), indicating areas needing improvement.

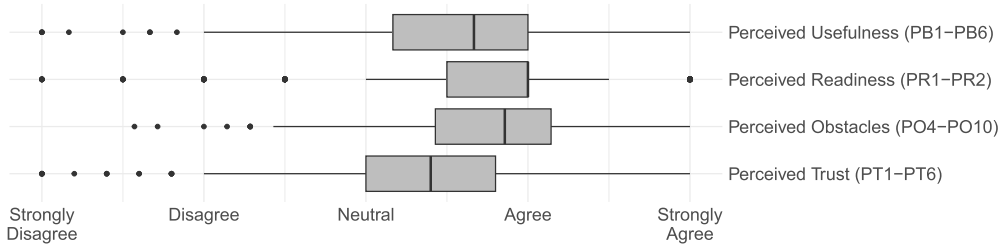

Fig. 4. Distribution of students' scores within each factor. The scores per factor were calculated by averaging over all items associated with the factor (Tabachnick et al., 2007).

Table 8
Multiple regressions results. Standard errors (in brackets) are heteroskedasticity-robust. The number of observations for all regressions was 665. Statistical significance levels: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Predictor	*Perceived Usefulness*	*Perceived Trust*	*Perceived Obstacles*	*Perceived Readiness*
(Intercept)	3.69(0.25)***	3.49(0.37)***	3.62(0.21)***	3.24(0.36)***
Age	-0.00(0.01)	-0.01(0.02)	0.01(0.01)	0.02(0.02)
Gender: Male	0.10(0.06)	0.01(0.05)	-0.15(0.05)***	0.12(0.07)
Program: Eng	-0.35(0.07)***	-0.07(0.07)	-0.04(0.06)	-0.06(0.09)
Program: M&P	-0.27(0.07)***	-0.00(0.07)	0.01(0.06)	0.00(0.09)
Level: Master	0.14(0.08)	0.02(0.10)	0.17(0.08)*	0.19(0.11)
R^2	8%	1%	5%	2%

In a final analysis, given the heterogeneity of the participating student populations, we investigated whether there were any differences in perception between students of different demographics. For the gender comparisons, we defined students identifying as women and non-binary as Minorities, as the statistical comparison with non-binary gender alone was impossible due to the small number of students ($N = 7$) identifying as non-binary. For the study level comparisons, we considered all Bachelor students as one group, as statistical comparisons for second year Bachelor were impossible due to the small number of participants ($N = 9$). We fit four multiple linear regression models, one for each emerged factor *Perceived Usefulness*, *Perceived Obstacles*, *Perceived Readiness*, and *Perceived Trust*), with student demographics as predictors (age, gender, level of studies, and program of studies). As there was no evidence of multicollinearity (we found variance inflation factor (VIF) scores < 3 for each predictor variable), we included all the predictors in the models. Table 8 lists the intercept and coefficients for each predictor variable and factor. The low R^2 values indicate that a substantial portion of the variability in shaping attitudes toward AI-EdTech remains unexplained by the demographic variables. However, the intercepts for all models were significant ($p < .001$), indicating baseline effects for each factor. Notably, gender significantly impacted perceived risks and obstacles: male participants expressed lower risks associated with AI-EdTech adoption than minorities ($p < .001$). Students in the Engineering, Mathematics, and Physics programs reported lower perceived usefulness of AI-EdTech ($p < .001$) than Computer Science and Communication Systems students. Master's level students showed a small increase in perceived obstacles ($p < .05$). Age, however, was not found to be significantly associated with any of the factors studied.

6.2. Qualitative analysis

Our survey also included two supplementary open-ended questions to get insights into AI's perceived benefits (´In your opinion, are there other benefits of using AI in the field of education not mentioned above?") and mistrust (´In your opinion, are there other reasons for not trusting AI in the field of education not mentioned above?") in educational settings beyond those captured by the Likert-scale items. 175 students answered at least one of the above questions.

108 students (16%) responded to the open-ended question about the additional benefits of AI in education. The benefits mentioned by students included AI's role in personalizing learning experiences ($N = 27$), AI's ability to enhance learner autonomy and reduce reliance on educators ($N = 7$), and AI's potential to augment teaching effectiveness ($N = 8$). The most frequently mentioned advantage ($N = 39$), involved AI's capacity to streamline educational processes, such as summarizing material, providing quick access to resources, performing spell checks, and translating texts. While the aforementioned advantages were already encapsulated within the existing Likert-scale items, novel insights emerged when students identified AI's contribution to creativity, citing its utility in fostering new ideas, broadening perspectives, and introducing innovative learning methodologies ($N = 22$). Example responses include *"[AI can] find new ideas, open the mind to what exists,"* and *"[AI can be] an inspiration for artistic work."* A further critical aspect ($N = 7$) was AI's support in managing workloads and maintaining mental health. For example, one student specified, "The AI can also analyze our mental health, etc., to warn us or a medical expert to avoid burn-out, depression, etc. by warning them that such a person needs help."

103 students (15%) offered insights into the factors diminishing trust in AI. Most of these responses highlighted concerns about AI-generated insights' accuracy and potential bias ($N = 52$). Additionally, students raised ethical concerns ($N = 33$), including the misuse of AI by third parties, privacy issues, and plagiarism. These concerns align with issues previously identified in the Likert-scale items. Notably, the analysis revealed an emergent theme not covered by the existing items: the risk of over-reliance on AI ($N = 20$), potentially eroding students' critical and creative thinking skills and their learning abilities in general. One participant, for example, wrote, *"[AI] might decrease our ability to reason and memorize, and our critical thinking."*. Other participants mentioned, *"The loss of wanting to understand for yourself,"* and *"[AI] Prevents one's own reasoning."*

Based on these insights, the next version of the survey will include new specific items that address AI's creativity as an additional benefit and over-reliance on AI as a possible risk involved in using AI in education.

7. Discussion

This study introduced a new instrument to evaluate the role of trust in students' perceptions and attitudes toward adopting AI-EdTech, supported by rigorous statistical validation to ensure its internal consistency. By applying EFA and CFA to responses from 665 STEM students, we identified four key factors shaping their willingness to adopt AI-EdTech: *Perceived Usefulness*, *Perceived Obstacles*, *Perceived Trust*, and *Perceived Readiness*. Our SEM analysis confirmed that trust positively influences the perceived usefulness of AI-EdTech, both directly and through readiness to use it. Moreover, drawing on prior TAM-based research on AI adoption in educational settings—which established perceived usefulness as a key driver of student acceptance—we argue that trust plays a pivotal role in shaping students' engagement with AI technologies (Foroughi et al., 2023; Celik & Muukkonen, 2023; Maheshwari, 2024; Strzelecki & ElArabawy, 2024). In our study, while students recognize the potential benefits of AI-EdTech, their trust remains near neutral, underscoring the opportunity to enhance their perceptions of AI-EdTech's benefits by fostering greater trust which in turn will shift their attitudes toward a more positive stance.

Our findings substantiate the considerable ethics-related concerns among students. Intriguingly, our study reveals that as students' knowledge about the internal workings of AI increases, so do their concerns. This is particularly evident when comparing Master's students in Computer and Communication Sciences with their Bachelor counterparts in all STEM disciplines. This finding suggests that increasing AI knowledge alone is insufficient to alleviate ethical concerns.

While the above discussions focus on the case where AI technologies are underutilized due to uncertainties and ethical concerns, there also exists the significant problem of misuse, where AI is incorrectly or inappropriately used (Glikson & Woolley, 2020; Lee & See, 2004). Misuse can stem from overtrust and overreliance on AI systems, a phenomenon where users place excessive confidence in AI's capabilities, assuming these systems are infallible or suitable for all types of decision-making. Our study's analysis of responses to open-ended questions emphasized students' concerns about overtrust that might lead to neglecting necessary human oversight. Thus, while fostering trust in AI is essential, ensuring a responsible use of AI is equally important.

7.1. Implications for practitioners

Below, we discuss several strategies that should be promoted by AI-EdTech developers, decision-makers, and practitioners to calibrate student trust in order to prevent both lack of trust and overreliance on AI in educational settings.

7.1.1. Promoting AI literacy

One clear way to decrease uncertainty is to increase knowledge. In the context of AI-EdTech, knowledge includes a distinct set of competencies predominantly centered around AI literacy and AI readiness (Laupichler et al., 2022; Long & Magerko, 2020; Luckin et al., 2022; Ng et al., 2021). In our study, 74% of students (based on item PR4 responses, see Fig. A.5) agreed that effectively utilizing AI-based tools for personalizing their learning will necessitate acquiring new skills, exemplified in item PR4 by an awareness of potential AI errors and biases and the critical and ethical consumption of AI-generated insights. These findings demonstrate that students recognize the importance of developing competencies associated with AI literacy. Such recognition may enhance their motivation to deepen their understanding of the fundamentals of AI and its capabilities and restrictions. Developing a curriculum explicitly dedicated to improving students' AI literacy is essential to address this educational need effectively (UNESCO et al., 2024b). This curriculum should address technical aspects of AI and critical thinking skills that enable students to evaluate AI technologies. Elements of the curriculum could include hands-on projects where students interact with AI systems and analyze their outputs, theoretical modules that explain the mechanisms of AI decision-making, and ethical discussions on the implications of AI in society. By integrating these components, the educational program would aim to build a comprehensive understanding of AI, equipping students with the knowledge and skills necessary to navigate and leverage this technology safely and effectively in their future careers. At the same time, educating teachers (e.g., through a similar AI-literacy curriculum) about AI's limitations can ensure that they remain active participants in AI-driven activities (UNESCO et al., 2024a).

7.1.2. Following ethics standards

Ethics-related issues are frequently overlooked in learning analytics and educational technology discussions (Alwahaby et al., 2022). This might be due to the fact that ethical concerns are hard to evaluate and tackle as they consist of a spectrum of diverse issues, including data privacy, security, lack of transparency, and the potential for biases and errors in AI algorithms. A promising approach to address the complex ethical challenges is grounded in the Theory of Procedural Justice (Roberson & Stewart, 2006; Kizilcec, 2016). Procedural justice describes how people's sense of fairness and, consequently, their trust are influenced not solely by the outcomes they receive but also profoundly by the quality and fairness of the *processes* that lead to these outcomes. Accordingly, the multifaceted ethical issues can be effectively addressed by establishing and following clear ethical procedures and guidelines that will be perceived by the end-users as fair and just. Echoing this need, Floridi and Cowls (2022) performed a comparative analysis of six prominent AI principles documents.[6] They outlined a framework comprising five fundamental principles for ethical AI: beneficence, non-maleficence, autonomy, justice, and explicability.[7]

7.1.3. Co-design of AI-EdTech

Additional student concerns reflected the emotional and psychological complexities inherent in human learning processes that AI systems do not easily acknowledge or understand. These concerns underline the necessity for AI-EdTech to be developed with a nuanced understanding of students' cognitive and emotional processes and with mechanisms to understand and adapt to diverse learning contexts and dynamics. Empirical evidence suggests that these concerns can be addressed by co-designing automated systems with students (Martinez-Maldonado et al., 2015). A recent literature review Sarmiento and Wise (2022) explored participatory design usage within the Learning Analytics community, focusing on participant groups, design process phases, and specific tools and techniques. The results highlight an increasing trend in these methods, especially in higher education with instructors and stakeholders. However, students were only involved in 29 of the 52 projects included in the review. Moreover, their primary contribution was often (25% of the 29 projects) limited to providing feedback through surveys, rendering them passive reviewers of the already designed products.

The results of our study emphasize the importance of including students in the design of AI-EdTech from very early stages, including the requirement settings.

7.1.4. Human-in-the-loop approach

On par with the ethical justifications of promoting autonomous end-user engagement with AI-EdTech (Holmes et al., 2022; Dietvorst et al., 2018), the existing corpus of studies on human-computer interaction (Nazaretsky et al., 2022a), as well as our results (based on item PT4 responses, see Fig. A.5), emphasize the importance of preserving end-user autonomy. This entails the need for AI-EdTech to equip both students

[6] The Asilomar AI Principles, the Montreal Declaration for Responsible AI, Ethically Aligned Design by IEEE, Ethical Principles by the European Commission's European Group on Ethics, the AI principles in the UK House of Lords report, and the Tenets of the Partnership on AI.

[7] In the context of AI, the term explicability closely relates to AI transparency, interpretability, and accountability.

and instructors with personalized, contextual adaptation capabilities and the ability to review AI-generated recommendations. In addition to reducing ethical concerns, providing such overwriting capabilities to instructors is a promising direction in treating the issues related to algorithm aversion (Dietvorst et al., 2015, 2018). Similarly to (Nazaretsky et al., 2022b, 2024; Gaube et al., 2021), we also found (based on item PT6 responses, see Fig. A.5) a prevailing preference for human advice over AI-generated recommendations, reflecting a deep-rooted inclination towards human interaction and judgment. This inclination stresses the need to design and implement so-called human-in-the-loop AI-EdTech, which aims to complement human expertise rather than replace it. Such an amalgam of human and AI competencies allows AI-EdTech to consider a broader learning context outside of online systems, including social, emotional, and psychological complexities involved in the learning process, as well as prevent the propagation of possible AI mistakes and inconsistencies to the students. On par with the above, recent work (Shneiderman, 2020) has introduced a bidimensional Human-Centered AI (HCAI) framework guiding AI designers and developers in creating AI-EdTech characterized by the composition of pronounced human oversight and advanced computational automation, in opposite to a more conservative consideration of unidimensional AI-autonomy vs. AI-automation taxonomy (Parasuraman et al., 2000).

7.2. Implications for researchers

In this subsection, we discuss how developing a robust trust instrument facilitates longitudinal analyses to capture the dynamic nature of trust, evaluate pedagogical interventions, and enhance methodological rigor supporting cumulative research on the role of trust in AI-EdTech adoption.

7.2.1. Facilitating longitudinal analyses

Research on trust in technology has shown that trust evolves as users compare their initial expectations with the actual performance of technology and gain experience over time. Studies grounded in Cognitive Dissonance Theory and Expectation Disconfirmation Theory (EDT) reveal that both positive and negative disconfirmations (unmet expectations) can *negatively* affect trust (Lankton et al., 2016). These findings highlight trust development's non-linear and complex nature, emphasizing the need to explore trust trajectories that consider the interplay between expectations, experiences, and outcomes over time.

Our robust trust instrument enables a comprehensive investigation of these dynamics across different stages of AI-EdTech adoption. It captures how trust is built, diminished, or restored in response to specific pedagogical interventions while accounting for students' initial and evolving perceptions of AI-EdTech trustworthiness. By integrating EDT with our instrument, we can gain deeper insights into the evolution of trust and develop evidence-based strategies to promote long-term trust and acceptance of AI-driven educational technologies.

7.2.2. Facilitating uniformity and reproducibility

Considering all the aforementioned trust-related concerns, promoting a more consistent and uniform multidimensional trust assessment instrument is imperative. Trust is a complex construct influenced by various cognitive, emotional, and contextual factors, making it critical to employ robust methodologies that capture these dimensions comprehensively. Our empirically validated trust instrument represents a significant step toward advancing methodological rigor in trust-related research within the AI-EdTech adoption process. By offering a standardized instrument for measuring trust in educational settings, this tool addresses critical challenges, including the lack of consistency in operational definitions, variability in measurement approaches, and the absence of cross-study comparability. As a result, it not only enhances the reproducibility of findings but also enables researchers to build on a shared foundation, fostering cumulative knowledge in the field. Moreover, this uniformity can facilitate meta-analyses and systematic reviews, essential for synthesizing evidence and drawing generalizable insights.

8. Limitations and future work

Validating a perceptions instrument is an iterative process that requires multifaceted verification (Butz & Branchaw, 2020). Initially, the instrument may not include enough items to capture the full range of perceptions. Additionally, new factors that emerge during validation may not be adequately represented, and the initial testing might be limited to specific populations. In the current instrument version, the AI-EdTech readiness factor is represented by a few items, and we hence plan to extend it in the next version of the instrument. In addition, we intend to incorporate new items that explore how creativity facilitated by AI can enhance learning benefits. Similarly, we will include items that assess potential risks associated with over-reliance on AI, reflecting the complexities and challenges this dependence may introduce. Furthermore, we aim to develop a dedicated dimension focusing specifically on AI literacy to more thoroughly gauge students' skills and competencies required to manage and utilize AI technologies effectively. These enhancements will allow for a more comprehensive and nuanced assessment of students' perceptions and attitudes toward AI in educational settings. Moreover, while our current validation focused on students with STEM backgrounds, our next step targets a broader student demographic.

9. Conclusions

This paper presents a novel instrument designed to assess students' trust in adopting AI-EdTech, along with a robust statistical validation of its internal consistency, ensuring that the instrument accurately captures the multifaceted nature of trust. Using EFA and CFA on responses from 665 higher education STEM students, we identified four critical factors shaping AI-EdTech adoption: two related to the perceived trustworthiness of AI-EdTech - *Perceived Usefulness* and *Perceived Obstacles* - and two reflecting student characteristics -*Perceived Trust* and *Perceived Readiness*. Our SEM analysis highlighted the crucial role of trust in enhancing the perceived usefulness of AI-EdTech, both directly and indirectly through the readiness to use AI-EdTech. Based on prior research of educational AI adoption grounded in the TAM, we assume that perceived usefulness significantly boosts user acceptance and usage intentions, thereby positioning trust as a critical determinant in students' adoption of AI-EdTech for learning.

Our contribution is twofold. First, we provide a standardized, validated instrument for evaluating trust in AI-EdTech, setting a foundation for more consistent and reliable future research. Second, we offer actionable insights into factors shaping AI-EdTech adoption, guiding the development of effective design and implementation strategies that address students' concerns and promote wider acceptance of these technologies.

CRediT authorship contribution statement

Tanya Nazaretsky: Writing – review & editing, Visualization, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Paola Mejia-Domenzain:** Writing – review & editing, Investigation. **Vinitra Swamy:** Writing – review & editing, Investigation. **Jibril Frej:** Writing – review & editing, Investigation. **Tanja Käser:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Open data, ethics

The study was approved by an ethical committee with ID: HREC 017-2023. Informed consent was obtained from all participants, and their privacy rights were strictly observed.

The data from this study are not available online due to privacy issues.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Table A.9
5-factor EFA model: standardized loadings (pattern matrix) based upon the correlation matrix. Loadings that are greater than 0.30 are highlighted in bold. $h2$ denotes the amount of variance in the item explained by the factors. It is the sum of the squared loadings (communality). $u2 = 1 - h2$ is residual variance (uniqueness). com denotes the item complexity, which measures how much an item reflects a single factor. The percentage of variance explained by each factor is indicated in the last row. The factors are ordered by the proportion of explained variance (decreasing order).

	F2	F1	F3	F5	F4	h2	u2	com
PB1	0.02	**0.40**	0.25	-0.01	-0.05	0.30	0.70	1.7
PB2	0.07	**0.52**	0.16	0.10	-0.11	0.38	0.62	1.4
PB3	0.01	**0.42**	0.23	0.15	-0.09	0.34	0.66	2.0
PB4	-0.03	**0.60**	-0.03	-0.07	0.19	0.42	0.58	1.2
PB5	-0.08	**0.40**	0.04	-0.04	0.17	0.24	0.76	1.5
PB6	0.01	**0.68**	0.03	0.00	0.00	0.49	0.51	1.0
PB7	-0.04	0.26	0.07	-0.04	0.12	0.12	0.88	1.7
PO1	**0.33**	0.10	-0.30	0.06	0.05	0.16	0.84	2.3
PO2	0.18	0.08	**-0.37**	-0.08	0.06	0.15	0.85	1.7
PO3	0.18	-0.07	0.01	-0.14	0.14	0.07	0.93	3.2
PO4	**0.49**	0.16	-0.03	-0.08	-0.02	0.28	0.72	1.3
PO5	**0.60**	0.01	-0.07	-0.04	0.04	0.37	0.63	1.0
PO6	**0.49**	-0.07	0.06	-0.10	0.07	0.27	0.73	1.2
PO7	**0.58**	0.12	-0.07	0.08	0.00	0.34	0.66	1.2
PO8	**0.67**	-0.09	0.07	0.09	0.01	0.45	0.55	1.1
PO9	**0.62**	-0.03	0.08	-0.03	-0.09	0.40	0.60	1.1
PO10	**0.45**	-0.02	0.05	-0.19	-0.01	0.26	0.74	1.4
PR1	0.07	0.04	**0.72**	0.04	0.08	0.60	0.40	1.1
PR2	0.01	0.16	**0.64**	0.00	0.05	0.56	0.44	1.1
PR3	0.15	0.14	-0.30	0.19	0.17	0.15	0.85	3.5
PR4	0.20	0.11	0.00	-0.08	0.20	0.11	0.89	2.8
PT1	-0.14	0.11	0.16	0.21	**0.35**	0.33	0.67	2.7
PT2	0.12	-0.12	0.20	-0.03	**0.49**	0.29	0.71	1.6
PT3	-0.06	0.07	0.02	0.05	**0.60**	0.42	0.58	1.1
PT4	0.06	0.08	-0.02	0.09	**0.44**	0.24	0.76	1.2
PT5	0.02	-0.01	0.05	**0.86**	0.01	0.74	0.26	1.0
PT6	-0.04	0.00	-0.06	**0.75**	0.02	0.56	0.44	1.0
Explained Variance	9.3%	7.6%	6.1%	5.9%	4.6%			

Table A.10
5-factor EFA model: correlations between factors. Correlations that are greater than 0.30 are highlighted in bold.

	F2	F1	F3	F5	F4
F2	-				
F1	0.08	-			
F3	0.10	**0.44**	-		
F5	-0.12	0.18	0.19	-	
F4	0.05	**0.31**	0.22	0.23	-

Table A.11
4-factor CFA models: factor loadings with Chronbach's α reliability values. All loadings are significant ($p < .001$).

Item	Initial Model	Final Model
Perceived Usefulness	$\alpha = 0.73$	$\alpha = 0.73$
PB1: AI can assist me in improving my responses (e.g., answers to open-ended questions, writing essays, etc.) by suggesting personalized feedback in real time.	0.58	0.59
PB2: AI can assist me in self-learning by adjusting automatically individualized learning tasks adapted to my personal needs and level of knowledge.	0.65	0.65
PB3: AI (e.g., ChatGPT) can serve me as a learning pal or teaching assistant.	0.53	0.53
PB4: AI can help teachers diagnose my strengths and difficulties so teachers can assign individualized learning tasks adapted to my personal needs and level of knowledge.	0.65	0.64
PB5: AI can assist me with managing my learning activities (e.g., identifying if I'm on track with my tasks, offering task prioritization, optimizing time management, etc.)	0.63	0.63
PB6: AI can assist me in preparing for exams by identifying my knowledge gaps and providing relevant learning materials (e.g., explanations, tasks, etc.)	0.64	0.64
Perceived Obstacles	$\alpha = 0.66$	$\alpha = 0.66$
PO4: AI's decision-making and reasoning processes are not transparent.	0.46	0.34
PO5: The origins, collection methods, and limitations of the data utilized by AI are not clear (e.g., who created the dataset, how the data was collected, etc.)	0.55	0.41
PO6: It's unclear who holds accountability for decisions made by AI.	0.37	0.30
PO7: The intentions and funding sources of AI-based technology providers are not transparent.	0.55	0.57
PO8: It's unclear when and what type of data AI gathers about me.	0.66	0.70
PO9: The use of data raises risks to my privacy (e.g., data breaches, surveillance and profiling, etc.)	0.59	0.65
PO10: The use of data raises ethical issues (e.g., possible biases, lack of diversity, AI mistakes, etc.)	0.46	0.49
Perceived Readiness	$\alpha = 0.78$	$\alpha = 0.78$
PR1: I believe I will be successful in using AI-based tools to personalize my learning when and if they are available.	0.88	0.88
PR2: I will use AI-based tools for personalizing my learning when and if they are available.	0.83	0.82
Perceived Trust	$\alpha = 0.66$	$\alpha = 0.68$
PT1: As AI-based tools for personalized learning become more prevalent, I will trust them more.	0.57	0.61
PT2: The more I know how an AI-based tool makes decisions (e.g., the steps involved in machine learning and the practices and challenges that each step entails, etc.), the more I can trust it.	0.57	0.39
PT3: The more data is available for AI-based tools, the more I can trust its insights.	0.68	0.48
PT4: The more autonomy AI-based tools provide me/my teacher to check/overwrite its automated decisions, the more I can trust them.	0.67	0.49
PT5: I rely on the recommendation of an AI-based tool at least as much as I trust a recommendation from a fellow learner.	0.57	0.51
PT6: I rely on the recommendation of an AI-based tool at least as much as I trust a recommendation from my teacher.	0.34	-

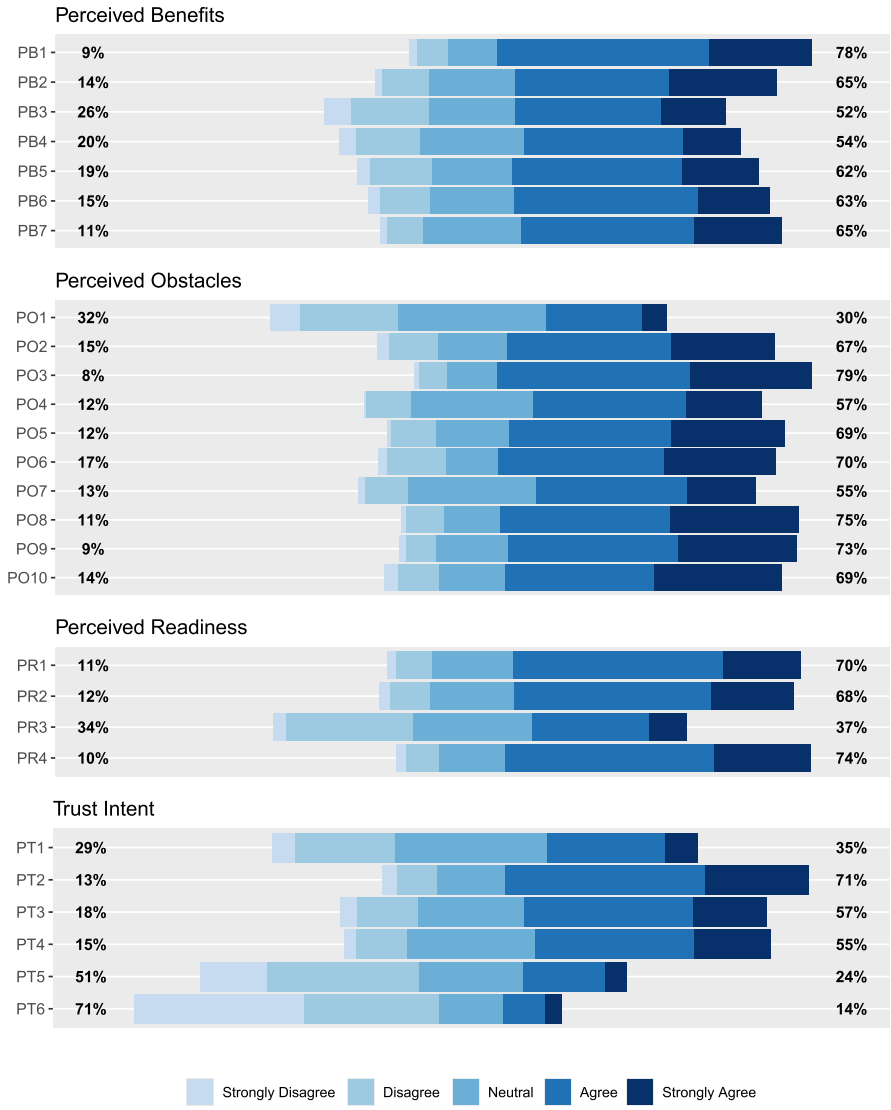

Fig. A.5. Proportion of responses for each Likert level across different items. Items are ordered based on the order of presentation.

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